

Paul Walko

Denver, CO | +1 (502) 509-6242 | paul@bigcavemaps.com | paul.walko.org | github.com/paulwalko

EMPLOYMENT

Senior DevOps Engineer

Aug. 2019 – Present

Ozmo, Inc (Associate → Senior, Jan. 2023)

Blacksburg, VA

- **AWS Architecture:** Architect production applications (ECS/Fargate, Lambda, RDS, OpenSearch) for maintainability and security; lead quarterly DevOps planning as most senior engineer on team
- **SOC 2:** Own end-to-end compliance program – IAM access reviews, vulnerability management, CI/change controls, audit logging, encryption, backup/DR testing, vendor risk reviews; liaison with external auditor
- **Observability:** Built ELK + Grafana stack ingesting ~1M log events/hour; primary system for production alerting and incident response
- **CI/CD:** Leading migration from Jenkins to GitHub Actions; embed with engineering teams on pipeline design and AWS CDK adoption
- **OpenVPN Cloud:** Implemented always-on split-tunnel VPN supporting remote work
- **Search Service:** Contributed to event-driven global search service in AWS
ozmo.com/blog/building-a-world-wide-real-time-search-service

Systems Engineer

Jan. – July 2019

Solers, Inc

Arlington, VA

- Worked with Docker Swarm and Ansible
- Deployed central logging solution for clustered environment

Cyber Security Engineer

May 2017 – Aug. 2018

Cisco

Raleigh-Durham, NC

- Developed Linux security module anti-malware for IoT devices
- Hardened Linux systems and kernels
- Analyzed and prevented potential security bugs

Research Assistant

Sep. 2016 – Dec. 2017

Hume Center at Virginia Tech

Blacksburg, VA

SKILLS

AWS: API Gateway, CloudFormation, CloudFront, CloudWatch, EC2, ECS/Fargate, ElasticSearch/OpenSearch, Lambda, RDS, Route53, S3

GIS Tools: QGIS, GDAL, Leaflet, GeoServer, MapProxy

Languages: Bash, C, Java, JavaScript/NodeJS, Python, Ruby

Tools: Ansible, AWS CDK, Docker, GitHub Actions, Google Workspace, Git, Grafana, Helm, Jenkins, Kubernetes, Librato, Linux, NewRelic, Tailscale, Ubiquiti

Other: Amateur Radio (KN4VDC), Argentine Tango (assist with lessons)

PROJECTS

Self-Hosted Services

- **Kubernetes Cluster:** IPv6-only multi-node k8s cluster on Tailscale running Browsertrix, Forgejo, Immich, Nextcloud, RustFS
- **Cavepedia:** github.com/cavepedia/cavepediav2. Semantic search + chat over decades of NSS caving documents in partnership with the National Speleological Society; pgvector embeddings, FastMCP server, PydanticAI agent, Auth0 RBAC.

- **Mergin Maps:** maps.caving.dev. Used on remote caving expeditions to manage GIS.
- **Mastodon:** social.bigcavemaps.com
- **Geoserver/Leaflet/MapProxy:** Used to analyze lidar and find potential caves.

FOSDEM 2026 Air Quality Monitoring

Feb. 2026

- Co-built distributed CO₂/temp/humidity sensor network deployed across 20+ devrooms (Meshtastic mesh, MQTT, Postgres, Grafana); results shared with FOSDEM staff for ventilation planning. paul.walko.org/blog/2026-02-07_fosdem-part2

Free/Open Software Contributions

- **2013 – Present:** Use and contribute to OpenStreetMap, especially not well known trails.
- **2019 – Present:** Archive Team – Help archive Internet services and manage massive amounts of data.

ORGANIZATIONS

VPI Cave Club | vpicave.club (Student Grotto of the NSS)

2020 - Present

- **Survey/Cartography:** Survey and map unexplored cave passage, including data entry and processing to create a finished map.
- **Training:** Teach new members basic caving skills, including vertical competence when climbing or abseiling rope.
- **Events:** Help organize regional caving events with 200+ attendees.

Virginia Tech Linux/Unix Users Group (VTLUUG) | vthuug.org, github.com/vthuug 2016 - Present

- **FreeIPA/LDAP:** Maintain accounts for email, shell access, and webserver.
- **Rebuilt Infrastructure:** Updated services to deploy with Ansible and run all services with Docker.
- **Organized funds:** Obtained \$1300 for new servers and equipment; oversaw purchasing additional components.

Outdoor Club at Virginia Tech (OCVT) | ocvt.club, github.com/ocvt

2017 - Present

- **Dolabra:** Built backend API in Golang and SQLite + front-end in plain HTML/JavaScript. Used Open-ID Connect for auth, Stripe for payments, and Docker for testing and deployment in AWS.
- **Appalachian Trail Maintenance:** Regularly clear trail of trees, brush, and clean shelters.

TALKS

Cave Rescue Communications (Toorcamp 2024): archive.is/iX4s8

Austrian Expedition to the Loser Plateau (NSS Convention 2023)

Newberry's Cave Exploration (NSS Convention 2022)

EDUCATION

Virginia Tech

Bachelor of Science in Computer Science; Minor in Cyber Security

Blacksburg, VA

Aug. 2015 – Dec. 2018